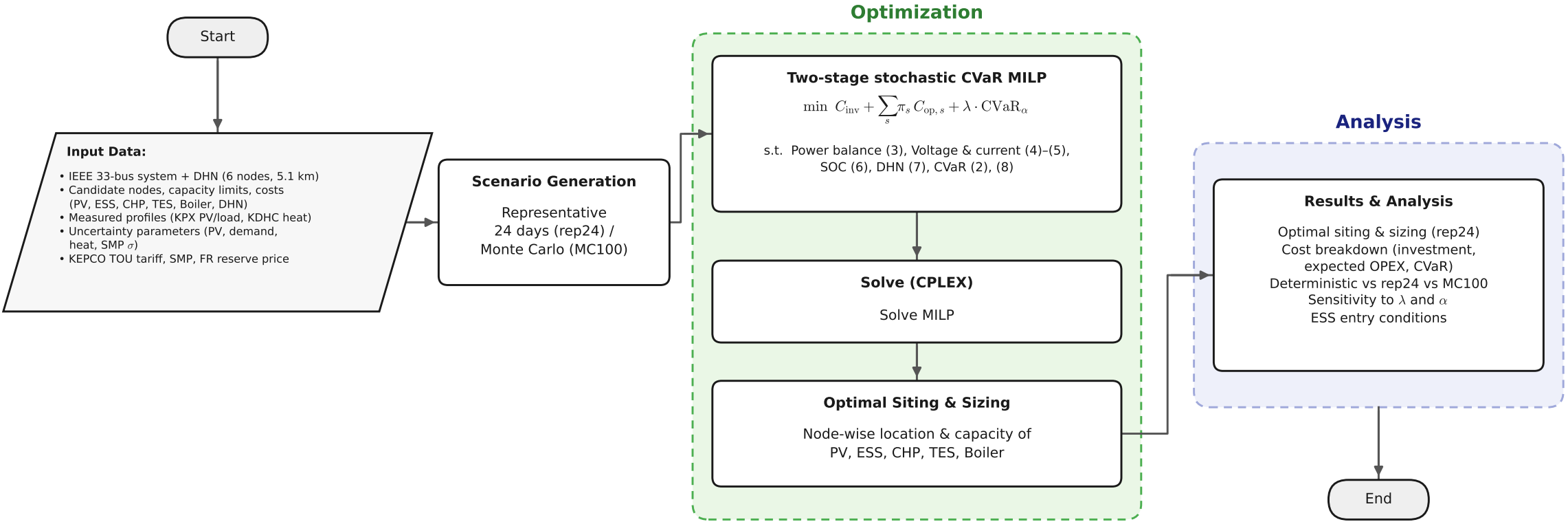
Start
Input Data:
• IEEE 33-bus system + DHN (6 nodes, 5.1 km)
• Candidate nodes, capacity limits, costs (PV, ESS, CHP, TES, Boiler, DHN)
• Measured profiles (KPX PV/load, KDHC heat)
• Uncertainty parameters (PV, demand, heat, SMP σ)
• KEPCO TOU tariff, SMP, FR reserve price
Scenario Generation
Representative 24 days (rep24) / Monte Carlo (MC100)
Optimization
Two-stage stochastic CVaR MILP
$\min\ C_{\text{inv}} + \sum_s \pi_s C_{\text{op},s} + \lambda \cdot \text{CVaR}_\alpha$
s.t. Power balance (3), Voltage & current (4)–(5), SOC (6), DHN (7), CVaR (2), (8)
Solve (CPLEX)
Solve MILP
Optimal Siting & Sizing
Node-wise location & capacity of PV, ESS, CHP, TES, Boiler
Analysis
Results & Analysis
Optimal siting & sizing (rep24)
Cost breakdown (investment, expected OPEX, CVaR)
Deterministic vs rep24 vs MC100
Sensitivity to λ and α
ESS entry conditions
End